

# Overview of Agentic AI Systems

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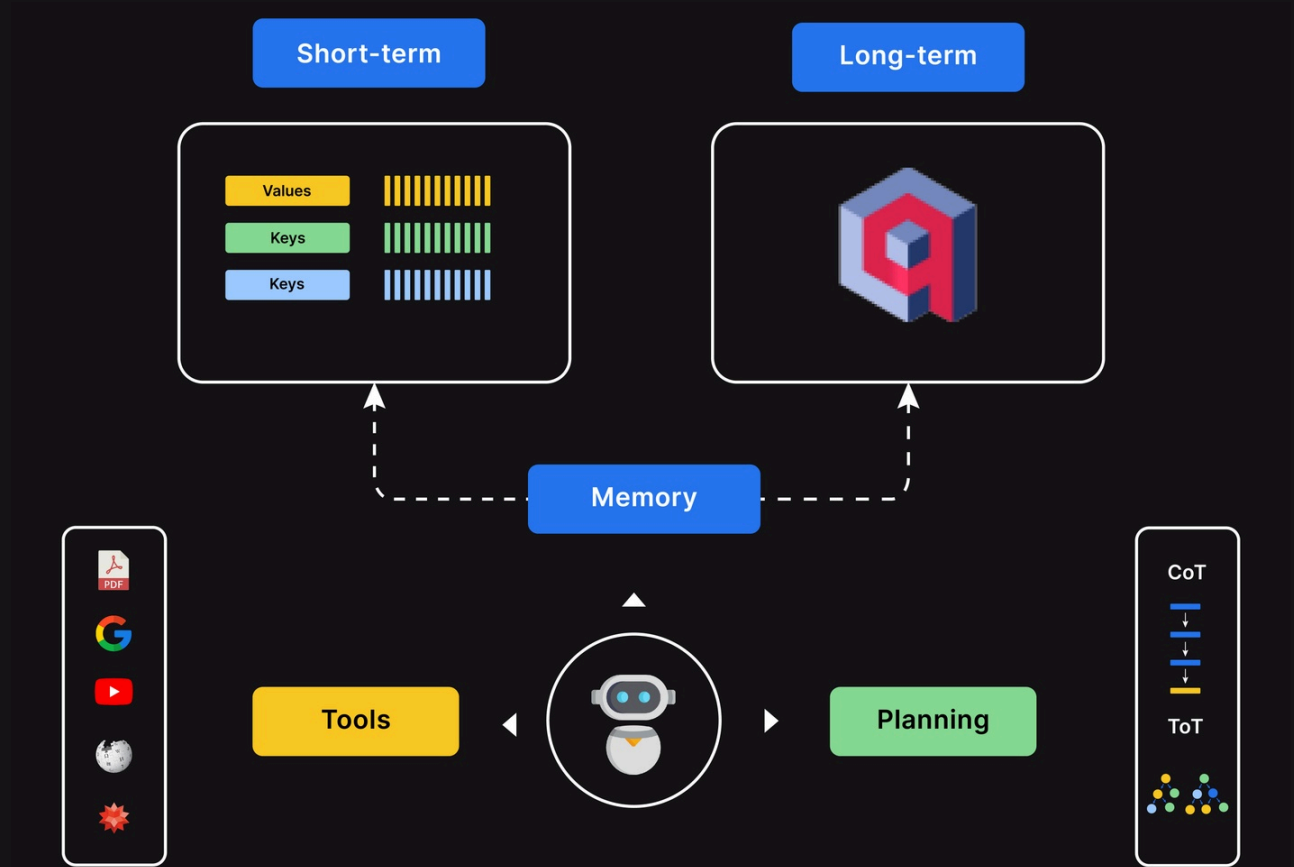
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# What is an Agent?

- **LLM as the Brain**
  - The agent's core, enabling understanding, reasoning, and response generation.
- **Tooling**
  - Connects to external tools for enhanced functionality like retrieving data or automating actions.
- **Reasoning / Planning routines**
  - Executes tasks logically with a structured decision-making.
- **Memory Components**
  - Retains temporary task data (short-term) and persistent knowledge (long-term) for adaptability.

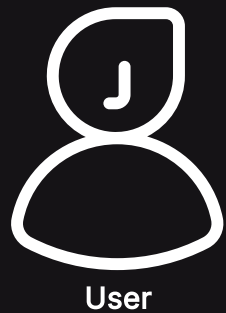


Source: <https://theneuralmaze.substack.com/>

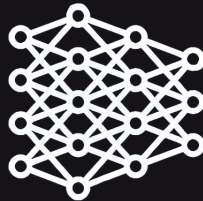
# LLM as the Agent's Brain



# Tools for Accessing the Outside World



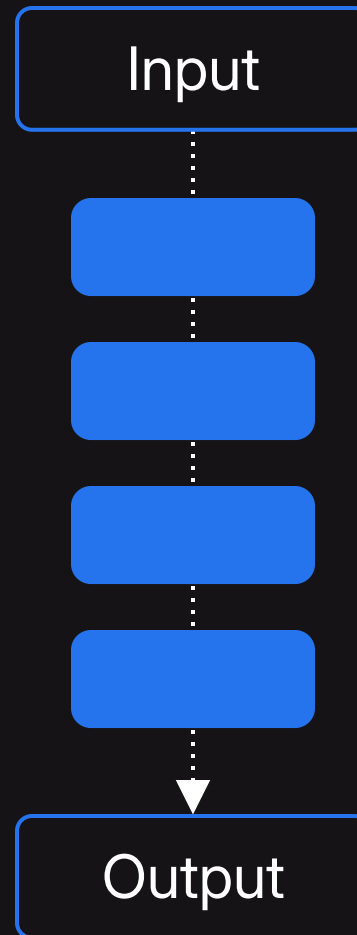
What is the weather in Hawaii?



```
AIMessage(  
  tool_calls=[  
    {  
      name: "get_weather"  
      args: {  
        location:  
"Hawaii",  
      },  
      id: "call_abc123",  
      type: "tool_call"  
    }  
  ]  
)
```

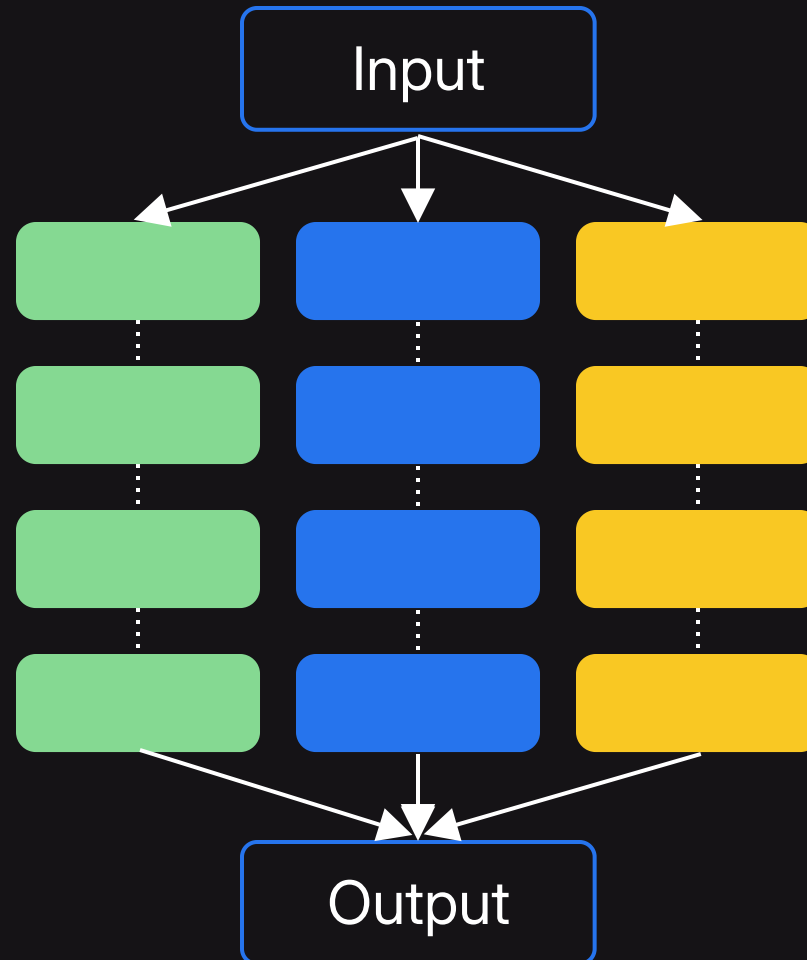
# Planning and Reasoning Heuristics

## CoT - Chain Of Thought



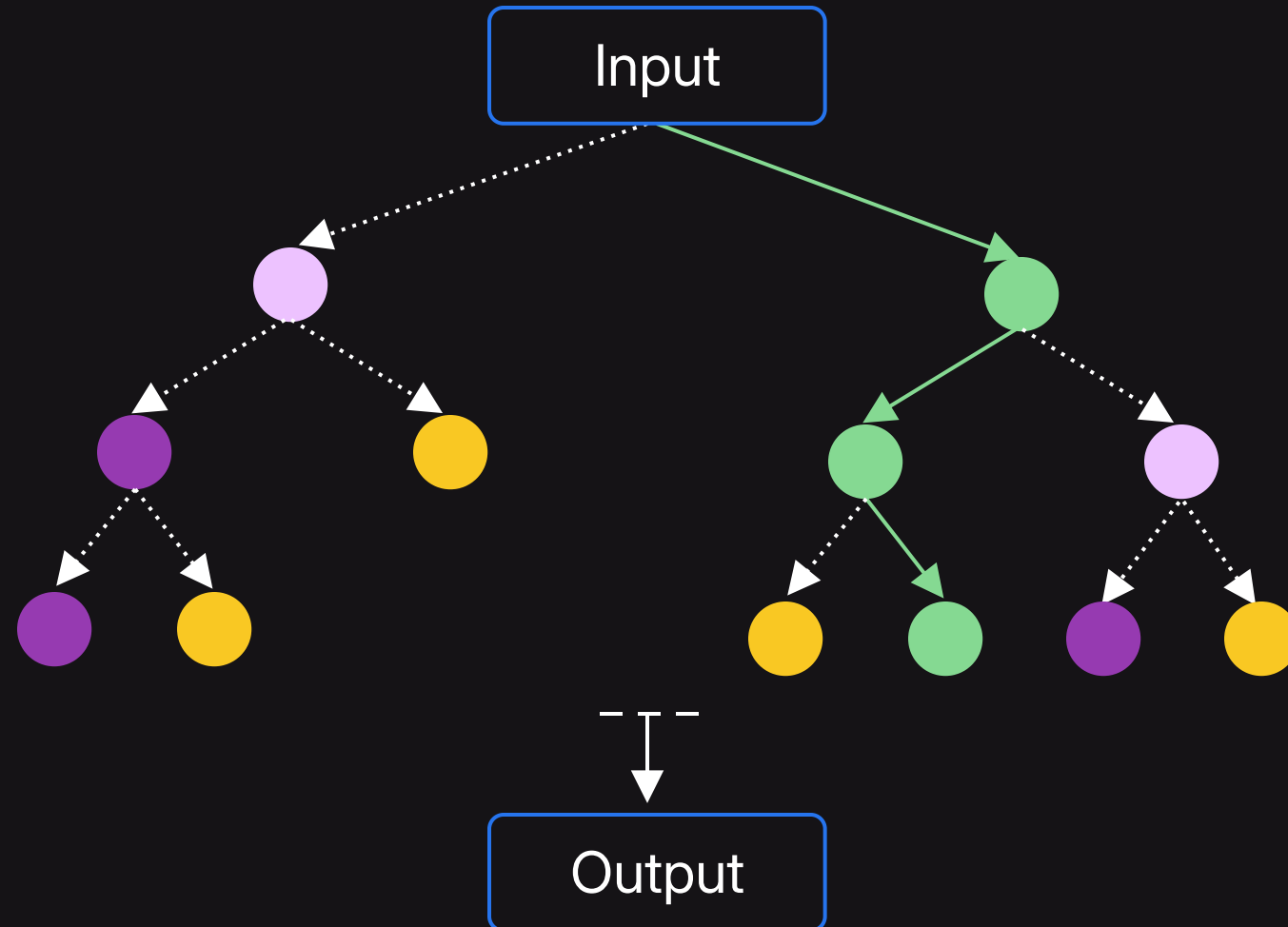
# Planning and Reasoning Heuristics

## COT-SC - Chain of Thought Self Consistency



# Planning and Reasoning Heuristics

## ToT - Three of Thoughts



# Planning and Reasoning Heuristics – ReAct

- ReAct is the most famous Reasoning technique.
- Consists of a loop of **Thought – Action – Observation**
- Every framework, nowadays, uses ReAct under the hood.

## (1d) ReAct (Reason + Act)

**Thought 1:** I need to search Apple Remote and find the program it was originally designed to interact with.

**Act 1:** `Search[Apple Remote]`

**Obs 1:** The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the `Front Row media center` program ...

**Thought 2:** Apple Remote was originally designed to control the `Front Row media center` program. I need to search `Front Row` next and find what other device can control it.

**Act 2:** `Search[Front Row]`

**Obs 2:** Could not find `[Front Row]`. Similar: `['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]`

**Thought 3:** `Front Row` is not found. I need to search `Front Row (software)` .

**Act 3:** `Search[Front Row (software)]`

**Obs 3:** `Front Row` is a discontinued media center software ...

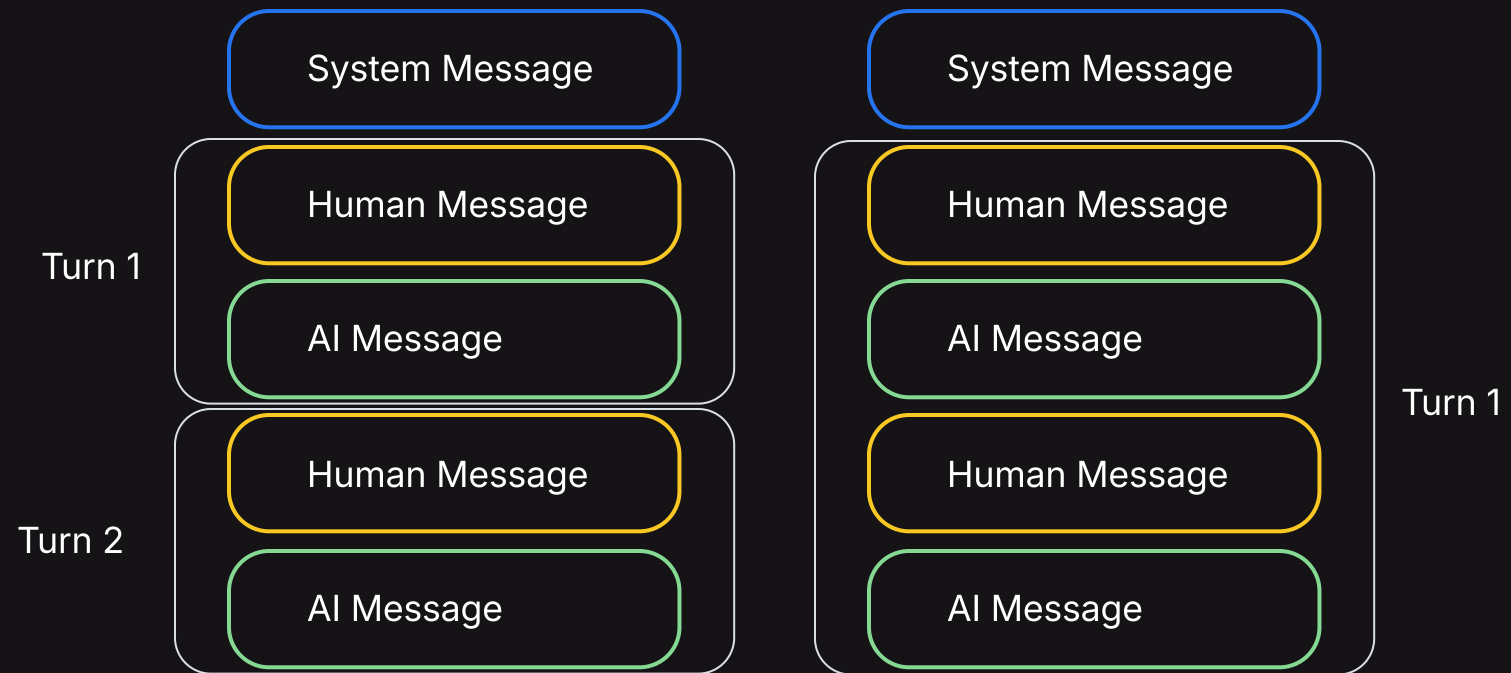
**Thought 4:** `Front Row (software)` is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

**Act 4:** `Finish[keyboard function keys]`

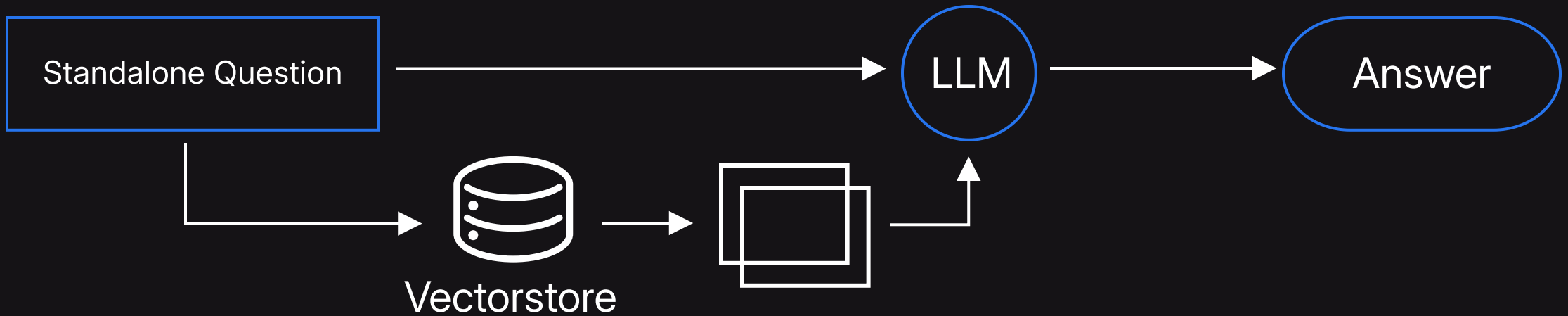
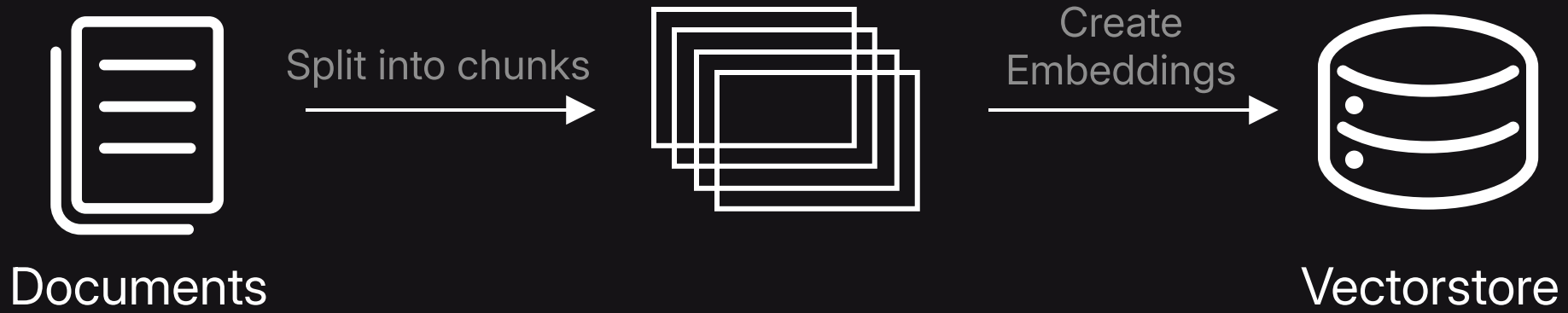


# Memory (short-term)

- Component necessary for applications with conversational interfaces.
- It allows access to a window of previous messages. We can think of it as short-term memory.
- Special caution must be taken with the size of this window to avoid exceeding the context limit of the LLM.



# Memory



**Thanks**